

TELESUPPORT HUB

Cloud-Based Telecom Support System Using Flask on AWS

SMARTINTERNZ COURSE PROJECT REPORT

PROJECT SUBMITTED BY

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Academic Year: 2025–2026

PROJECT TITLE

TeleSupport Hub: Cloud-Based Telecom Support System Using Flask on AWS

COURSE PLATFORM

SmartInternz

TECHNOLOGY STACK

Frontend Technologies

- React.js
- Bootstrap 5.3
- HTML5
- CSS3
- JavaScript
- Axios
- React Icons

Backend Technologies

- Python
- Flask Framework
- Flask-CORS
- REST APIs
- JWT Authentication

Database Technologies

- MySQL Database

Cloud Technologies

- AWS EC2
- AWS RDS
- AWS Cloud Infrastructure

Version Control

- Git
- GitHub

CERTIFICATE

This is to certify that Mr. Fardin Firoj Atapade (PRN No: 2022011031003), student of Dr. D. Y. Patil Agriculture And Technical University, Talsande, has successfully completed the project entitled "TeleSupport Hub: Cloud-Based Telecom Support System Using Flask on AWS" as part of the SmartInternz Course Program.

The project demonstrates practical implementation of Cloud Computing, Full Stack Web Development, Database Management, Authentication Systems, REST API Development, and Cloud Deployment using AWS technologies.

DECLARATION

I hereby declare that the project entitled "TeleSupport Hub: Cloud-Based Telecom Support System Using Flask on AWS" submitted by me is my original work completed as part of the SmartInternz Course Program.

I further declare that this project has not been submitted elsewhere for the award of any degree, diploma, certificate, or qualification.

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to SmartInternz for providing an excellent learning platform and practical exposure to Full Stack Web Development and Cloud Computing concepts.

I am also thankful to Dr. D. Y. Patil Agriculture And Technical University, Talsande, for providing the academic support and resources required for successful completion of this project.

Finally, I would like to thank all mentors, faculty members, friends, and family members who directly or indirectly contributed to the successful completion of this project.

ABSTRACT

TeleSupport Hub is a cloud-based telecom customer support platform designed to streamline customer issue management and improve support efficiency.

The system enables telecom customers to register, authenticate securely, raise support tickets, track issue status, and communicate with support teams. Administrators can monitor tickets, update statuses, and manage customer requests through a centralized dashboard.

The frontend is developed using React.js and Bootstrap 5.3, while the backend is developed using Flask and REST APIs. MySQL is used for database management, and AWS cloud services are utilized for hosting and deployment.

The system provides a scalable, reliable, and secure solution for telecom customer support management.

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1. INTRODUCTION

Telecommunication companies receive thousands of customer complaints every day regarding network issues, recharge failures, SIM card problems, internet connectivity issues, and calling-related concerns.

Managing these complaints manually often leads to delayed resolutions and poor customer experience.

TeleSupport Hub provides a centralized digital platform that allows customers to report issues and track their status in real time while enabling administrators to efficiently manage support operations.

2. PROBLEM STATEMENT

Traditional telecom support systems suffer from several challenges:

- Manual complaint handling.
- Lack of real-time tracking.
- Delayed customer response.
- Poor complaint management.
- Limited transparency.
- Inefficient communication between users and support teams.

These challenges negatively impact customer satisfaction and operational efficiency.

3. PROJECT OBJECTIVES

The primary objectives of TeleSupport Hub are:

- Provide a centralized telecom support platform.
 - Enable online ticket creation.
 - Improve complaint resolution speed.
 - Maintain secure authentication.
 - Provide real-time ticket tracking.
 - Improve communication between customers and support teams.
 - Utilize cloud infrastructure for scalability.
 - Enhance overall customer satisfaction.
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4. EXISTING SYSTEM

The existing support system relies on:

- Phone calls
- Emails
- Manual ticket logs
- Spreadsheet management

Limitations:

- Time-consuming process.
 - Human errors.
 - No centralized dashboard.
 - Lack of status tracking.
 - Poor scalability.
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5. PROPOSED SYSTEM

TeleSupport Hub introduces:

- Secure User Registration
- User Authentication
- Ticket Management System
- Status Tracking
- Administrative Dashboard
- Cloud-Based Infrastructure
- Centralized Database Management

Benefits:

- Faster issue resolution.
 - Better transparency.
 - Improved scalability.
 - Enhanced customer experience.
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6. SYSTEM REQUIREMENTS

Hardware Requirements

- Processor: Intel i3 or above
- RAM: 4GB or above
- Storage: 20GB Free Space

- Internet Connection

Software Requirements

- Windows/Linux
 - VS Code
 - Python
 - Flask
 - Node.js
 - React.js
 - MySQL
 - Git
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7. TECHNOLOGY STACK

Frontend

- React.js
- Bootstrap 5.3
- HTML5
- CSS3
- JavaScript
- Axios

Backend

- Flask
- Python
- JWT Authentication
- Flask-CORS

Database

- MySQL

Cloud

- AWS EC2
 - AWS RDS
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8. SYSTEM ARCHITECTURE

Customer



React Frontend



REST API Communication



Flask Backend



JWT Authentication



MySQL Database



Admin Dashboard

9. DATABASE DESIGN

Users Table

Field	Type
id	Integer
full_name	Varchar
email	Varchar
password	Text
created_at	Timestamp

Tickets Table

Field	Type
id	Integer
user_id	Integer
full_name	Varchar
email	Varchar

Field	Type
phone	Varchar
issue_type	Varchar
subject	Varchar
description	Text
status	Enum
created_at	Timestamp
updated_at	Timestamp

10. API DESIGN

Authentication APIs

POST /api/auth/signup

POST /api/auth/login

POST /api/auth/send-otp

POST /api/auth/verify-otp

Ticket APIs

POST /api/tickets

GET /api/tickets

GET /api/tickets/{id}

PUT /api/tickets/{id}

DELETE /api/tickets/{id}

11. SECURITY IMPLEMENTATION

Security Features:

- JWT Authentication
- Password Hashing
- Secure API Endpoints
- Protected Routes
- Environment Variables
- Database Security

12. CLOUD DEPLOYMENT

Frontend Deployment

Hosted using AWS Cloud Services.

Backend Deployment

Flask application deployed on AWS EC2.

Database Deployment

MySQL Database deployed on AWS RDS.

Benefits:

- High Availability
 - Scalability
 - Reliability
 - Secure Infrastructure
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13. PROJECT WORKFLOW

1. User Registration
 2. Email Verification
 3. User Login
 4. Raise Ticket
 5. Store Ticket in Database
 6. Admin Reviews Ticket
 7. Update Status
 8. User Tracks Progress
 9. Issue Resolution
 10. Ticket Closure
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14. TESTING

The system was tested for:

- Registration Testing
- Login Testing
- Ticket Creation Testing
- Ticket Update Testing

- Database Testing
 - API Testing
 - Authentication Testing
 - Cloud Deployment Testing
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15. RESULTS

The developed system successfully:

- Handles customer complaints efficiently.
 - Provides secure authentication.
 - Maintains ticket records.
 - Tracks ticket progress.
 - Supports cloud deployment.
 - Improves telecom support operations.
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16. FUTURE ENHANCEMENTS

- AI-Based Ticket Classification
 - Chatbot Integration
 - Live Chat Support
 - SMS Notifications
 - Email Notifications
 - Analytics Dashboard
 - Mobile Application
 - Multi-Language Support
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17. LEARNING OUTCOMES

Through this project, the following skills were developed:

- Full Stack Development
- React.js Development
- Flask Backend Development
- REST API Development
- Database Management

- JWT Authentication
 - Cloud Deployment
 - AWS Services
 - Git & GitHub
 - Software Architecture Design
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18. CONCLUSION

TeleSupport Hub successfully demonstrates the implementation of a cloud-based telecom customer support platform capable of managing customer complaints efficiently.

The project integrates React.js frontend development, Flask backend services, MySQL database management, JWT authentication, and AWS cloud deployment to provide a scalable and secure telecom support solution.

The system improves customer satisfaction, enhances operational efficiency, and demonstrates practical application of modern cloud computing and full-stack development technologies.

19. REFERENCES

1. React.js Documentation
 2. Flask Documentation
 3. MySQL Documentation
 4. AWS EC2 Documentation
 5. AWS RDS Documentation
 6. SmartInternz Learning Resources
 7. JWT Documentation
 8. Bootstrap 5.3 Documentation
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THANK YOU

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SmartInternz Course Project

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